Summarized Paper Reports - Team 3

Paper 1: *Tourism Recommendation System Based on Semantic Clustering and Sentiment Analysis* (Preprint, 2020)

This paper addresses the limitations of traditional recommender systems, such as collaborative filtering (requiring many ratings) and content-based filtering (lacking personalization). The proposed solution combines semantic clustering of user reviews with sentiment analysis to extract deeper insights into tourist preferences.

Aspect	Details
Datasets Used	• **TripAdvisor Review Dataset** (Egypt + worldwide attractions) • Contains **tourist text reviews + metadata** • Preprocessing: stop-word removal, stemming, tokenization
Models & Methods Applied	• **Semantic Clustering**: k-means clustering on TF-IDF and Word2Vec embeddings to group reviews into themes (e.g., history, food, adventure) • **Sentiment Analysis**: polarity classification (positive, negative, neutral) for each cluster • **Matching Process**: user preference profile (clusters + sentiments) compared with attraction profiles to generate ranked recommendations
Results Produced	• **Precision**: **~81%** • **Recall**: **~76** % • **F1-score**: **~78** % • Outperformed baseline collaborative filtering models significantly
Limitations & Future Work	• Dataset limited to TripAdvisor reviews (not global scale) • Language dependency: trained only on English reviews • **Cold-start problem** for new users with no review history • No context-awareness (time, season, group size not considered) **Future Work:** Expand dataset to multilingual reviews (Arabic + English), integrate temporal/contextual factors, include multimodal data (photos, check-ins, ratings), build a real-time mobile app prototype

Paper 2: *Optimisation Using Natural Language Processing: Personalized Tour Recommendation for Museums* (arXiv:1501.01252, Jan 2015)

This paper introduces a personalized tour recommendation system tailored for museum visits. It integrates Natural Language Processing (NLP) with optimization methods to maximize visitor satisfaction by balancing intrinsic importance of artworks with user preferences, within time constraints.

Aspect	Details
Datasets Used	• Musée de l'Orangerie (Paris) dataset • **144 artworks** across **10 rooms** • Data fields: artist, title, year, textual description
Models & Methods Applied	• **NLP (Enertex)**: Textual energy method to score "intrinsic importance" of artworks from descriptions • **Visitor Preference Modeling**: user interests represented as vectors (e.g., preferred artists), similarity measured with cosine similarity • **Optimization (ILP – Integer Linear Programming)**: models museum layout as graph, finds optimal tour path under time limits • **Scoring Functions Tested**: f1 = equal scores, f2 = Enertex-based, f3 = visitor preference-based, f4 = hybrid (weighted Enertex + preference)
Results Produced	• Evaluation metric: **Relevance Percentage (rp)** (how well tours match user preferences) • **Hybrid model (f4)** improved tour relevance by up to **40%** compared to baseline (f1) • Optimal tours designed for **~150 minutes (2.5 hours)** showed the highest visitor satisfaction • f2 (Enertex) worked best for discovery-oriented visitors, f3 (preference-based) best for strongly biased tastes, and f4 (hybrid) best overall
Limitations & Future Work	• Tested only on a **single museum dataset** (limited scalability) • Visitor preferences restricted to **artists only**, not broader themes/styles • No contextual factors (e.g., group dynamics, accessibility, real-time changes) **Future Work:** Extend personalization to themes/styles, integrate advanced NLP (topic modeling, sentiment analysis), develop **mobile/voice-guided tours**, and scale optimization for larger museums with thousands of artworks

Paper 3: A Deep Learning Approach to Ancient Egyptian Hieroglyphs Classification (https://tinyurl.com/4hxewyvd, 2021)

This paper investigates the automatic classification of Ancient Egyptian hieroglyphs using **Convolutional Neural Networks (CNNs)**. thousands of symbols , high variability ,degraded inscriptions **. The goal is to evaluate how CNNs perform against traditional methods and to introduce a custom model** (**Glyphnet**) for hieroglyph recognition

Aspect	Details
Datasets Used	• **A public dataset** extracted from the Pyramid of Unas, containing 4310 labeled grayscale images of hieroglyphs, annotated according to Gardiner's sign list. This dataset offers a standardized benchmark with relatively clean images. • **A custom dataset** compiled by the authors from multiple supports (stone, papyrus, etc.), reflecting more realistic conditions with noise and erosion. This dataset was chosen to test the models' robustness in challenging, real-world scenarios. ***However, it is important to note that neither of these datasets is publicly available. The first one, extracted from the Pyramid of Unas, has not been released beyond the scope of the paper, while the second was privately compiled by the authors from various supports and remains inaccessible.*** As a practical alternative, researchers may refer to publicly accessible datasets such as the Egyptian Hieroglyphs dataset on Hugging Face , which provides a comparable collection of labeled hieroglyph images. Although not identical, this dataset can serve as a useful proxy for experimentation and benchmarking.
Models & Methods Applied	**• ResNet-50: Developed by Microsoft Research with 23M+ parameters. • Uses residual blocks to solve vanishing gradient issues. • Processes 224x224 images.** **• Inception-v3: Developed by Google with 42 layers Uses inception modules to capture features at multiple scales. • Processes 299x299 images** **• Xception: Evolution of Inception model using depthwise separable • More efficient with 299x299 images.** **• Glyphnet (Proposed Network): Inspired by Xception, but tailored for hieroglyphs Uses separable convolutions, batch normalization, dropout, and global average pooling. Only ~0.5M parameters compared to 20M+ in others Faster, lighter, less prone to overfitting.**

<table>
<tr>
<td>Results Produced</td>
<td>
TABLE 1. Networks performance using the transfer learning approach
<table>
<tr><th>Architecture</th><th>Accuracy</th><th>Precision</th><th>Recall</th><th>F1-Score</th></tr>
<tr><td>ResNet-50</td><td>0.906</td><td>0.882</td><td>0.825</td><td>0.840</td></tr>
<tr><td>Inception-v3</td><td>0.864</td><td>0.730</td><td>0.737</td><td>0.717</td></tr>
<tr><td>Xception</td><td>0.834</td><td>0.715</td><td>0.720</td><td>0.703</td></tr>
</table>
TABLE 2. Networks performance using training from scratch
<table>
<tr><th>Architecture</th><th>Accuracy</th><th>Precision</th><th>Recall</th><th>F1-Score</th></tr>
<tr><td>ResNet-50</td><td>0.945</td><td>0.919</td><td>0.905</td><td>0.903</td></tr>
<tr><td>Inception-v3</td><td>0.948</td><td>0.917</td><td>0.904</td><td>0.900</td></tr>
<tr><td>Xception</td><td>0.956</td><td>0.919</td><td>0.930</td><td>0.919</td></tr>
<tr><td>Glyphnet</td><td>0.976</td><td>0.975</td><td>0.965</td><td>0.968</td></tr>
</table>
</td>
</tr>
<tr>
<td>Contributions of the Work</td>
<td>
• Applied deep learning methods to hieroglyph classification, a novel area in Egyptology.

• Compared multiple CNN architectures on two datasets of hieroglyph images.

• Proposed Glyphnet , a new CNN optimized for this task.

• Demonstrated superior performance of Glyphnet in bothaccuracy and speed.

• Laid foundations for future AI-based tools in automatic recognition, classification, and translation of hieroglyphs
</td>
</tr>
<tr>
<td>Drawbacks & Limitations</td>
<td>
• Limited datasets: Only 4309 images across 40 categories, relatively small compared to large-scale benchmarks.

• Class imbalance: Some hieroglyph categories had far fewer images.

• Computational resources: Although Glyphnet is lighter, CNN training still requires GPUs.

• Focus on single-character classification: Full text recognition and translation remain future challenges.
</td>
</tr>
</table>

Future Directions	•Extending classification from single hieroglyphs to full words and connected text. •Integrating the classifier into a full OCR pipeline that can support transliteration and translation. •Enhancing robustness against damaged or eroded inscriptions. •Leveraging larger linguistic and image datasets for broader Egyptological applications.

Paper 4: A BERT-Based Intelligent Chatbot for Tourism Information and Recommendation in Vietnam (Published, 2023)

This paper presents the development of a full-stack mobile application featuring an intelligent chatbot designed to assist foreign tourists in Vietnam. The system addresses the challenge of information accessibility by using advanced natural language processing (NLP) to understand user queries, provide destination information, and offer personalized recommendations, thereby significantly enhancing the tourist experience.

.

Aspect	Details
Datasets Used	• **TWCS (Customer Support on Twitter) Dataset**: A large corpus of customer service tweets. [Available on Kaggle] • **Tourism Subset**: A processed, hand-labeled subset of 7,319 dialogue samples extracted from airline Twitter feeds (American Air, British Airways, Southwest Air) relevant to travel inquiries. • **Labels**: 11 user intents (e.g., greet, goodbye, get_suggest, give_city, give_interests).

	• **Preprocessing**: Data was split for 5-fold cross-validation (80% training, 20% testing).
Models & Methods Applied	• **Intent Classification**: A comparative analysis of three NLP models: Bag-of-Words, BERT [Original Paper], and RoBERTa [Original Paper]. • **Chatbot Framework**: Built using the **RASA** open-source framework [RASA Official Site] for dialogue management. • **System Architecture**: A comprehensive architecture featuring a Flutter/Dart mobile app, a React-based content management web app, a **PostgreSQL** database [PostgreSQL Official Site], and RASA NLU/action servers. • **Deployment**: The trained BERT model was integrated into a production-ready Android application.
Results Produced	• **BERT** achieved the highest performance after 200 training epochs: - **Accuracy**: **90.0%** - **F1-Score**: **90.1%** • The deployed chatbot application successfully handled predefined scenarios for greetings, information retrieval, and place recommendations with high accuracy. • The system significantly outperformed the Bag-of-Words baseline and the RoBERTa model on this specific intent classification task.
Limitations & Future Work	• **Dataset Limitations**: Trained on a domain-specific subset of customer service data, which may not encompass the full range of tourist queries or conversational nuances. • **Language Dependency**: The system is designed only for English, limiting its utility for the large number of non-English speaking tourists in Vietnam. • **Context-Awareness**: Does not incorporate real-time contextual factors like weather, time, or dynamic local events. **Future Work**: Expand the training dataset with more diverse tourist intents and Vietnamese language support, integrate real-time contextual and multimodal data (e.g., images, real-time reviews), and enhance the personalization of the recommendation engine.

Paper 5: Image Based Hieroglyphic Character Recognition (Published, 2018)

This paper presents an image-based Optical Character Recognition (OCR) system designed to automatically recognize ancient Egyptian hieroglyphs and translate them into English. The proposed framework tackles challenges of segmentation, reading order preservation, and character matching using advanced image processing and feature extraction techniques. The goal is to support tourists, researchers, and Egyptologists by providing an accessible tool for understanding hieroglyphic inscriptions.

Aspect	Details
Datasets Used	• A dataset of about **700 query images** was used. • It covered **197 different hieroglyphic characters**, organized according to Gardiner's codes. • The dataset also included different orientations (left/right) for some symbols to support correct recognition.
Models & Methods Applied	• OCR-based pipeline using **Image Processing**. • **Image Acquisition**: grayscale conversion, Canny edge detection, Hough Transform. • **Segmentation**: bounding box + connected components labeling, preserving reading order (top-to-bottom, right-to-left when applicable). • **Post-processing**: noise removal, resizing, filtering out small images. • **Feature Extraction & Matching**: Histogram of Oriented Gradients (HOG) + Euclidean distance. • **English Mapping**: Gardiner's codes mapped to English translations. • Alternative techniques tested: Template Matching, SURF (lower accuracy).
Results Produced	• **Segmentation accuracy**: avg. 84.33% compared to manual segmentation. • **Recognition accuracy**: avg. 66.64%. • Outperformed state-of-the-art Chinese character recognition methods applied to hieroglyphs (55.27%). • Produced meaningful English translations, though not grammatically structured.

Limitations & Future Work	• **Limitations**: – Translated English lacks proper grammar/syntax. – Mainly supports vertical column inscriptions; not generalized to all hieroglyphic layouts. – Low-quality/blurry hieroglyphs reduce accuracy. • **Future Work**: – Integrate **NLP** for grammatical English translation. – Develop a **mobile application** for tourists/researchers. – Expand dataset for broader coverage of hieroglyphic formats.